

Quadratic Differentially 4-Uniform Permutations in Dimensions Divisible by Four: A Desarguesian Obstruction and Non-Existence in Dimension 8

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Abstract

Quadratic S-boxes are attractive for masked implementations because they admit three-share threshold implementations without decomposition into several stages, and differential uniformity 4 is the best possible value for a quadratic permutation of even dimension. By a result of Charpin and Peng, a quadratic differentially 4-uniform permutation of $\mathbb{F}_2^n$ is differentially two-valued and all its components have rank $n - 2$. All known examples of such permutations live in dimensions $n \equiv 2 \pmod{4}$. We study the case $4 \mid n$. We observe that the kernels of the derivatives of such a permutation form a line spread of $\mathbb{F}_2^n$ which is totally isotropic for all components and is mapped onto a line spread of the codomain. We show that this spread cannot be Desarguesian when $4 \mid n$; more generally, a quadratic permutation whose derivative kernels form a Desarguesian t -spread exists only if n/t is odd, which contains the classical fact that quadratic APN permutations exist only in odd dimension. As a consequence no $\mathbb{F}_4$ -quadratic map, for instance no map $x^{2^k+1} + \ell(x)$ with k even and ℓ linear, is a differentially 4-uniform permutation when $4 \mid n$ (for power maps alone this is classical). For $n = 8$ we settle the question: by an exhaustive computer search over the possible spaces of components, cross-checked by a second, separately written implementation, there is no quadratic differentially 4-uniform permutation of $\mathbb{F}_2^8$; more generally every quadratic $(8, 8)$ -function whose nonzero derivatives are all 4-to-1 has a bent component. The search rests on a simple closure property of the radicals of the components.

Keywords: differential uniformity, quadratic permutation, S-box, spread, alternating bilinear form, constant-rank subspace, computer-assisted proof.

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1 Introduction

Masked implementations of block ciphers pay for every non-linear operation. A threshold implementation of a function of algebraic degree d needs at least $d + 1$ shares [20]; for a quadratic S-box three shares suffice for a correct and non-complete sharing computed in one stage, whereas S-boxes of higher degree need more shares or a decomposition into several stages of lower degree separated by registers [2]. (Uniformity of the shared function is a separate requirement, which may need fresh randomness or a particular sharing; a low algebraic degree alone does not make an implementation secure.) So the cryptographic quality achievable by quadratic permutations is a natural design limit. For a permutation of $\mathbb{F}_2^n$ with n even the differential uniformity is at least 4 when the permutation is quadratic (Remark 2); the value 4 is attained by the Gold permutations x^{2^k+1} with $\gcd(k, n) = 2$ precisely when $n \equiv 2 \pmod{4}$ [10, 21], and other known quadratic examples, such as the binomials of Bracken, Tan and Tan [4] and the Dembowski–Ostrom quadrinomials over $\mathbb{F}_{2^{2m}}$ with m odd studied in [16], also live in dimensions $n \equiv 2 \pmod{4}$. For $n \leq 6$ all quadratic permutations are

classified up to affine equivalence [8]; for $n = 6$ eight of the classes have differential uniformity 4. For $4 \mid n$ differentially 4-uniform permutations do exist, for instance the inverse map [21] and the cubic power map $x^{2^{2k}+2^k+1}$ for $n = 4k$ with k odd [3], but no quadratic one is known. We show that none exists for $n = 8$.

The same dichotomy between $n \equiv 2$ and $n \equiv 0 \pmod{4}$ appears in the decomposition problem for permutations into quadratic ones [19]: if $4 \nmid n$, every permutation of $\mathbb{F}_{2^n}$ is a composition of quadratic power permutations and affine permutations $x \mapsto ax + b$, whereas if $4 \mid n$, no power permutation with differential uniformity below 16 is a composition of quadratic and linear power permutations [1]. Quadratic permutations with differential uniformity 4 in dimension $4 \mid n$ are therefore of interest both as candidate S-boxes and as building blocks of such decompositions.

Question. Does $\mathbb{F}_2^n$ with $4 \mid n$ admit a quadratic permutation with differential uniformity 4? The first open case was $n = 8$, the dimension of the AES S-box (which is based on the inverse map and is not quadratic); we answer it negatively (Theorem 22).

Known structure. Charpin and Peng [7, Theorems 1 and 7] proved that a quadratic differentially 4-uniform permutation of $\mathbb{F}_2^n$ (n even) is differentially two-valued $\{0, 4\}$, has no bent component and has nonlinearity $2^{n-1} - 2^{n/2}$ with a single Walsh amplitude; in the language of alternating forms, every component has rank exactly $n - 2$ and every derivative kernel has dimension 2 (Theorem 5 below). Their proof rests on the identity [7, Theorem 5] which we recall as Lemma 1.

Contributions.

1. Kernel spreads (Section 3): the derivative kernels of a quadratic DU-4 permutation form a line spread that is totally isotropic for all components; the restriction of the permutation to each kernel line is linear, and the images form a line spread of the codomain. This gives a partial normal form. (Isotropy of each single kernel is [18, Lemma 1].)
2. The Desarguesian obstruction (Section 4), our first main result: a quadratic permutation all of whose derivative kernels have dimension t and form a Desarguesian t -spread exists only if n/t is odd (Theorem 13). For $t = 1$ this is the classical fact that quadratic APN permutations exist only in odd dimension; for $t = 2$ it shows that, if $4 \mid n$, the kernel spread of a quadratic DU-4 permutation is not Desarguesian. This excludes all $\mathbb{F}_4$ -quadratic maps, for example Gold maps x^{2^k+1} with k even plus arbitrary linear maps (for the power maps themselves the conclusion is classical), and covers the case $n = 4$.
3. Non-existence for $n = 8$ (Section 5), our second main result: there is no quadratic DU-4 permutation of $\mathbb{F}_2^8$ (Theorem 22). By the result of Charpin and Peng it suffices to show that $\text{Alt}(8, \mathbb{F}_2)$ has no 8-dimensional subspace of constant rank 6 in which every point lies in exactly three radicals. We prove this by an exhaustive search built on a radical-closure lemma (Lemma 23), whose completeness is proved in Appendix A; the search takes a few CPU-hours and was cross-checked by a second, separately written implementation and a certificate for its symmetry reductions.
4. Local structure of kernel spreads (Section 6): for every even $n \geq 6$, two kernel lines span a 4-space containing 2, 3 or 5 kernel lines, and the dimension of $\langle \beta(L, M) \rangle$ is determined by this number (Proposition 24). We also summarise earlier partial results for $n = 8$ obtained with other methods, among them that the quadratic part of such a permutation could have no symmetry.

5. A necessary condition in terms of constant-rank subspaces of alternating forms, and a stronger conjecture that would settle the question (Section 7).

2 Preliminaries

Let $V = \mathbb{F}_2^n$. For $F : V \rightarrow V$ and $b \in V \setminus \{0\}$ we write $F_b(x) = \langle b, F(x) \rangle$ for the components of F . F is quadratic if its algebraic degree is at most 2, that is, if all its second-order derivatives are constant (affine maps are included); we normalise $F(0) = 0$ and write

$$\beta(u, v) = F(u + v) + F(u) + F(v),$$

a symmetric bi-additive map with $\beta(u, u) = 0$. For each $b \neq 0$, $B_b(u, v) = \langle b, \beta(u, v) \rangle$ is an alternating bilinear form with radical $R_b = \text{rad } B_b$; $\text{rank } B_b$ is even. For $a \neq 0$ put $K_a = \{v : \beta(a, v) = 0\} \ni a$ and $s_a = \dim K_a \geq 1$. Since $D_a F(x) = F(x + a) + F(x) = \beta(a, x) + F(a)$, the differential uniformity is $\delta_F = \max_{a \neq 0} 2^{s_a}$. A Boolean function is *bent* if its Walsh transform has constant absolute value $2^{n/2}$; a quadratic component is bent iff $\text{rank } B_b = n$, and semi-bent (Walsh values in $\{0, \pm 2^{(n+2)/2}\}$, n even) iff $\text{rank } B_b = n - 2$. Bent functions are never balanced [6], while every component of a permutation is balanced. Composing F with affine permutations on either side and adding affine maps (EA-equivalence) preserves δ_F and the ranks of the components; composing with affine permutations preserves bijectivity. When we identify V with $\mathbb{F}_{2^n}$ (Section 4), components are indexed through the absolute trace pairing: $F_b(x) = \text{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(bF(x))$ and $B_b(u, v) = \text{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(b\beta(u, v))$.

3 Structure of quadratic differentially 4-uniform permutations

Lemma 1 (Counting identity [7, Theorem 5]). *For every quadratic F ,*

$$\sum_{a \neq 0} (2^{s_a} - 1) = \sum_{b \neq 0} (2^{n - \text{rank } B_b} - 1).$$

Proof. Both sides count the pairs (a, b) of nonzero vectors with $B_b(a, \cdot) = 0$: for fixed a these b form $\{b : \langle b, \beta(a, v) \rangle = 0 \ \forall v\} = (\text{Im } \beta(a, \cdot))^\perp$, of dimension $n - (n - s_a) = s_a$; for fixed b these a form $R_b \setminus \{0\}$. $\square$

Remark 2. If n is even and F is a quadratic permutation then no component is bent (bent functions are not balanced), so $\text{rank } B_b \leq n - 2$ and $|R_b| \geq 4$ for all b ; Lemma 1 then gives $\sum_a (2^{s_a} - 1) \geq 3(2^n - 1)$, hence $\delta_F \geq 4$ (cf. [7, Theorem 1]).

Remark 3. The same double count applies to the restriction of β to $U \times U$ for any subspace U : $\sum_{a \in U \setminus \{0\}} (2^{n - \dim \beta(a, U)} - 1) = \sum_{b \neq 0} (2^{\dim U - \text{rank}(B_b|_U)} - 1)$.

Remark 4. In [7] the identity is written as $\sum_{\lambda \neq 0} 2^{\ell(\lambda)} = \sum_{a \neq 0} 2^{d(a)}$, where $\ell(\lambda) = \dim R_\lambda$ and $d(a) = s_a$; subtracting $2^n - 1$ from both sides gives the form above. For quadratic APN functions ($s_a = 1$) it gives the classical fact that in even dimension such functions have bent components [7, Remark 3].

The following theorem is due to Charpin and Peng [7, Theorem 1, Lemma 4 and Theorem 7]. We include the short derivation from Lemma 1 because the equality case is used repeatedly below.

Theorem 5 ([7]). *Let n be even and F a quadratic permutation of $\mathbb{F}_2^n$ with $\delta_F = 4$. Then $\text{rank } B_b = n - 2$ for every $b \neq 0$ and $s_a = 2$ for every $a \neq 0$.*

Proof. By Remark 2 the right-hand side of Lemma 1 is at least $3(2^n - 1)$, with equality iff all ranks equal $n - 2$; $\delta_F = 4$ gives $s_a \leq 2$, so the left-hand side is at most $3(2^n - 1)$, with equality iff all $s_a = 2$. $\square$

Remark 6. In the terminology of [7], F is differentially two-valued $\{0, 4\}$ and plateaued with single amplitude $2^{(n+2)/2}$ [7, Theorem 7]. For a quadratic function without bent components, Lemma 1 shows that the first property implies the second (every term on the right is at least 3 and the left-hand side is $3(2^n - 1)$), and that the second implies the first if $\delta_F \leq 4$. Whether the single amplitude alone forces $\delta_F = 4$ is the question discussed in Section 7. For general (not necessarily quadratic) plateaued functions, related relations between amplitudes and differential uniformity are studied in [17] (e.g. [17, Theorem 5.6]); we do not need them.

Remark 7. The permutation hypothesis is needed: x^5 on $\mathbb{F}_{2^8}$ is quadratic with $\delta = 4$ but has components of ranks 4 and 8.

Proposition 8 (Isotropic spread). *If $s_a = 2$ for all $a \neq 0$, then $\Sigma = \{K_a : a \neq 0\}$ is a line spread of V (a partition of $V \setminus \{0\}$ into 2-dimensional subspaces minus 0), and $\beta(L, L) = 0$ for every $L \in \Sigma$.*

Proof. If $u \in K_a \setminus \{0, a\}$ then $a \in K_u$ by symmetry of β , and $u \in K_u$, so $K_u \supseteq \langle a, u \rangle = K_a$ and equality holds by dimension. Hence every $x \neq 0$ lies in K_x , and if $x \neq 0$ lies in $K_a \cap K_b$ then $K_a = K_x = K_b$: the K_a partition $V \setminus \{0\}$. Finally, for $L = K_a$ and $y \in L \setminus \{0\}$ we have $K_y = L$, so $\beta(y, L) = 0$; thus $\beta(L, L) = 0$. $\square$

Remark 9. The isotropy of the kernels (elements of K_a are pairwise β -orthogonal) appears in [18, Lemma 1] for quadratic functions with $\delta_F = 4$; what is added here is the spread structure and its consequences below.

Corollary 10 (Spread correspondence). *Under the hypotheses of Theorem 5, the restriction $F|_L$ is $\mathbb{F}_2$ -linear for every $L \in \Sigma$, and $\{F(L) : L \in \Sigma\}$ is a line spread of the codomain.*

Proof. β vanishes on $L \times L$ and $F(0) = 0$, so $F|_L$ is additive; injectivity of F makes $F(L)$ a 2-dimensional subspace and distinct images meet only in 0. The images cover the codomain because F is bijective with $F(0) = 0$. $\square$

Lemma 11 (Partial normal form). *Under the hypotheses of Theorem 5 there are bases of the domain and the codomain such that $\beta(e_{2i}, e_{2i+1}) = 0$ for $0 \leq i < n/2$ and $\beta(e_0, e_j) = e_{j-2}$ for $2 \leq j \leq n-1$.*

Proof. Spread lines $L_1, \dots, L_{n/2}$ with $V = L_1 \oplus \dots \oplus L_{n/2}$ exist. Let U be a direct sum of $k/2$ spread lines ($k \leq n-2$, $\dim U = k$). A spread line either lies in U , or meets U in exactly one point and has two points outside U , or is skew to U ; and distinct spread lines meet U in distinct points. If no spread line were skew to U , every one of the $2^n - 2^k$ points outside U would lie on a spread line of the second kind, and there are at most $2^k - 1$ such lines, carrying at most $2(2^k - 1) < 2^n - 2^k$ outside points. So some spread line is skew to U , and induction gives the decomposition. Take e_{2i}, e_{2i+1} a basis of L_{i+1} . As $K_{e_0} = L_1 = \langle e_0, e_1 \rangle$, the linear map $\beta(e_0, \cdot)$ has kernel $\langle e_0, e_1 \rangle$ and rank $n-2$, so the vectors $\beta(e_0, e_j)$, $2 \leq j \leq n-1$, are linearly independent; compose F with a linear permutation of the codomain mapping them to $e_0, \dots, e_{n-3}$. $\square$

4 Desarguesian kernel spreads

A quadratic map is $\mathbb{F}_{2^t}$ -quadratic if its polar map β is $\mathbb{F}_{2^t}$ -bilinear for some structures of $\mathbb{F}_{2^t}$ -vector space on the domain and the codomain. For example, over $\mathbb{F}_{2^n}$ with $t \mid n$ every map

$\sum_{i < j} c_{ij} x^{2^{ti} + 2^{tj}} + \ell(x)$ with $c_{ij} \in \mathbb{F}_{2^n}$ and ℓ an $\mathbb{F}_2$ -linear map is $\mathbb{F}_{2^t}$ -quadratic; this includes the Gold maps x^{2^k+1} with $t \mid k$ plus arbitrary linear maps. All proofs in this section rest on the following elementary fact.

Lemma 12. *Let $E = \mathbb{F}_{2^t}$, let X be an E -vector space of dimension m and let $C : X \times X \rightarrow E$ be an alternating E -bilinear form. Then $B = \text{Tr}_{E/\mathbb{F}_2} \circ C$ is an alternating $\mathbb{F}_2$ -bilinear form with $\text{rad } B = \text{rad } C$. In particular $\dim_{\mathbb{F}_2} \text{rad } B = t(m - \text{rank}_E C)$, where $\text{rank}_E C$ is even.*

Proof. Clearly B is alternating and $\text{rad } C \subseteq \text{rad } B$. If $u \notin \text{rad } C$, then $C(u, \cdot)$ is a nonzero E -linear map $X \rightarrow E$, hence surjective; as $\text{Tr}_{E/\mathbb{F}_2}$ is not identically zero on E , there is y with $B(u, y) = \text{Tr}_{E/\mathbb{F}_2}(C(u, y)) \neq 0$, so $u \notin \text{rad } B$. Finally $\text{rad } C$ is an E -subspace of E -dimension $m - \text{rank}_E C$, and alternating forms have even rank. $\square$

Theorem 13. *Let F be a quadratic permutation of $\mathbb{F}_2^n$ with $s_a = t$ for every $a \neq 0$ (so $\delta_F = 2^t$ and F is differentially two-valued). Suppose that the kernels form a Desarguesian t -spread, that is, there is an $\mathbb{F}_{2^t}$ -vector space structure on $\mathbb{F}_2^n$ (in particular $t \mid n$) whose one-dimensional subspaces are the kernels: $K_x = \mathbb{F}_{2^t} x$ for every $x \neq 0$. Then n/t is odd.*

Proof. Write $\text{Tr} = \text{Tr}_{\mathbb{F}_{2^t}/\mathbb{F}_2}$ and $m = n/t$. Since $\lambda x \in K_x$ for $\lambda \in \mathbb{F}_{2^t}$, $\beta(x, \lambda x) = 0$; polarising gives $\beta(\lambda x, y) = \beta(x, \lambda y)$. Let $\{\theta_j\}$ be a basis of $\mathbb{F}_{2^t}/\mathbb{F}_2$ and $\{\theta_j^*\}$ its trace-dual basis, and put $\Phi(x, y) = \sum_j \beta(\theta_j x, y) \otimes \theta_j^* \in \mathbb{F}_2^n \otimes_{\mathbb{F}_2} \mathbb{F}_{2^t}$. Expanding $\theta_j \lambda = \sum_k \text{Tr}(\theta_j \lambda \theta_k^*) \theta_k$ gives

$$\Phi(\lambda x, y) = \sum_k \beta(\theta_k x, y) \otimes \sum_j \text{Tr}(\theta_j \lambda \theta_k^*) \theta_j^* = \sum_k \beta(\theta_k x, y) \otimes \lambda \theta_k^* = \lambda \Phi(x, y),$$

and $\Phi(x, \lambda y) = \Phi(\lambda x, y)$ by self-adjointness. Φ is symmetric (by self-adjointness and symmetry of β), $\Phi(x, x) = 0$, and $(\text{id} \otimes \text{Tr})\Phi(x, y) = \beta((\sum_j \text{Tr}(\theta_j^*) \theta_j)x, y) = \beta(x, y)$. Hence $B_b = \text{Tr} \circ C_b$ with $C_b = (\langle b, \cdot \rangle \otimes \text{id}) \circ \Phi$ an alternating $\mathbb{F}_{2^t}$ -form on $\mathbb{F}_{2^t}^m$, and by Lemma 12 $\dim_{\mathbb{F}_2} R_b = t(m - \text{rank}_{\mathbb{F}_{2^t}} C_b)$ with $\text{rank}_{\mathbb{F}_{2^t}} C_b$ even. If m is even these dimensions are multiples of $2t$; they are nonzero because a component with $R_b = 0$ would be bent, hence not balanced. Then Lemma 1 gives $(2^n - 1)(2^t - 1) = \sum_b (2^{\dim R_b} - 1) \geq (2^n - 1)(2^{2t} - 1)$, a contradiction. $\square$

Remark 14. For $t \geq 3$ the argument of Proposition 8, which uses $t = 2$, does not show that the kernels form a spread when $s_a = t$ for all a ; this is why the spread structure is part of the hypothesis.

Corollary 15. (i) *Every quadratic APN permutation of $\mathbb{F}_2^n$ has n odd.*

(ii) *Let $F = \sum \nu_r x^{2^i + 2^j}$ be a Dembowski–Ostrom polynomial over $\mathbb{F}_{2^n}$ all of whose exponents have $t \mid i$ and $t \mid j$, where $t \mid n$. If F is a differentially two-valued $\{0, 2^t\}$ permutation, then n/t is odd.*

Proof. (i) is the case $t = 1$: the kernel spread is the set of points. (ii) Here β is $\mathbb{F}_{2^t}$ -bilinear, so $\mathbb{F}_{2^t} a \subseteq K_a$, with equality since $s_a = t$. $\square$

Remark 16. Part (i) is classical, see e.g. [6]; the case $t = 2$ of Theorem 13 is Corollary 17 below. The binomials of [4, Theorem 1.2] over $\mathbb{F}_{2^{3k}}$ (see also [7, Theorem 9]) are of the form in (ii), which explains the hypothesis there that k/t is odd; for that family its necessity is proved directly in [4, Lemma 2.1], where $F(\gamma x) = F(x)$ for an element γ of order $2^t + 1$ when k/t is even. The quadrinomials of [4, Problem 5.1] carry the same hypothesis; Qu, Xiong and Li [22] showed, with the help of a computer, that they are differentially 4-uniform but need not be permutations. The condition of Theorem 13 cannot be weakened: the Gold permutations x^{2^k+1} with $\gcd(k, n) = t$ and n/t odd have Desarguesian kernel spreads.

Corollary 17 (Desarguesian obstruction). *Let $n \equiv 0 \pmod{4}$ and let F be a quadratic permutation of $\mathbb{F}_2^n$ with $\delta_F = 4$. Then the isotropic spread Σ of Proposition 8 is not Desarguesian.*

Proof. By Theorem 5, $s_a = 2$ for all $a \neq 0$. If Σ were Desarguesian, Theorem 13 with $t = 2$ would give that $n/2$ is odd. $\square$

Remark 18. For $t = 2$ the form Φ of the proof of Theorem 13 is explicit; we state this in the generality needed in Proposition 24. Let X be an $\mathbb{F}_2$ -space with an $\mathbb{F}_4$ -structure given by $\omega \in GL(X)$ with $\omega^2 + \omega + 1 = 0$, and let $w \in \mathbb{F}_4$ with $w^2 = w + 1$; the trace-dual basis of $\{1, w\}$ is $\{w^2, 1\}$. Let B be any alternating $\mathbb{F}_2$ -form on X with $B(x, \omega x) = 0$ for all x . Polarising gives $B(\omega x, y) = B(x, \omega y)$, and

$$C(x, y) = B(\omega x, y) + w^2 B(x, y) \in \mathbb{F}_4$$

satisfies $C(\omega x, y) = B(\omega^2 x, y) + w^2 B(\omega x, y) = w B(\omega x, y) + B(x, y) = w C(x, y)$ and $C(x, \omega y) = C(\omega x, y)$; it is symmetric with $C(x, x) = 0$, hence an alternating $\mathbb{F}_4$ -bilinear form, and $\text{Tr}_{\mathbb{F}_4/\mathbb{F}_2} \circ C = B$ since $\text{Tr}(1) = 0$ and $\text{Tr}(w^2) = 1$. By Lemma 12, $\text{rank}_{\mathbb{F}_2} B = 2 \text{rank}_{\mathbb{F}_4} C \equiv 0 \pmod{4}$.

Corollary 19. *Let $n = 2m$ with m even and let F be an $\mathbb{F}_4$ -quadratic map on $\mathbb{F}_2^n$. Then no component of F has rank $n - 2$. In particular no $\mathbb{F}_4$ -quadratic map, for instance no map $x^{2^k+1} + \ell(x)$ on $\mathbb{F}_{2^n}$ with k even and ℓ linear, is a permutation with $\delta_F = 4$.*

Proof. Every $\mathbb{F}_2$ -linear functional on an $\mathbb{F}_4$ -space is $\text{Tr}_{\mathbb{F}_4/\mathbb{F}_2} \circ \lambda$ for an $\mathbb{F}_4$ -linear functional λ . Hence $B_b = \text{Tr}_{\mathbb{F}_4/\mathbb{F}_2} \circ C_b$ with $C_b = \lambda_b \circ \beta$ alternating and $\mathbb{F}_4$ -bilinear, and by Lemma 12 $\text{rank}_{\mathbb{F}_2} B_b = 2 \text{rank}_{\mathbb{F}_4} C_b \equiv 0 \pmod{4}$, whereas $n - 2 \equiv 2 \pmod{4}$. The last statement follows from Theorem 5. $\square$

For power maps alone the last statement is classical: x^{2^k+1} is a permutation of $\mathbb{F}_{2^n}$ only if $n/\gcd(k, n)$ is odd, which for $4 \mid n$ forces $4 \mid \gcd(k, n)$ and hence $\delta = 2^{\gcd(k, n)} \geq 16$ [10, 21] (see also [1]). What is new is that adding a linear map, or passing to arbitrary $\mathbb{F}_4$ -quadratic maps or to Desarguesian kernel spreads, does not help.

Remark 20. For m odd the obstruction disappears and x^5 on $\mathbb{F}_{2^{2m}}$ is an example. Two results in the literature are close. For $\mathbb{F}_q$ -quadratic maps, [18, Theorem 5] proves $\delta_F \geq q$ (and computes the boomerang uniformity when equality holds for a permutation). For Dembowski–Ostrom polynomials $\sum \nu_r x^{2^i+2^j}$ with $t \mid i, t \mid j, t \mid n$ and without bent components, [7, Theorem 8] shows that F is differentially two-valued $\{0, 2^t\}$ if and only if every component has radical dimension exactly t (their proof observes that radicals are $\mathbb{F}_{2^t}$ -subspaces). Neither result contains a condition on n/t . The additional ingredient here is that the radical is the radical of an alternating $\mathbb{F}_4$ -form, so its $\mathbb{F}_4$ -codimension is even; this forces m to be odd, and Theorem 13 extends this from $\mathbb{F}_4$ -bilinear β to Desarguesian kernel spreads. We have not found this parity condition in the literature on differentially uniform functions or on constant-rank spaces of alternating forms (Section 7 discusses the latter).

Remark 21. Every line spread of $PG(3, 2)$ is regular (a regulus of $PG(3, 2)$ has only three lines), hence Desarguesian by the Bruck–Bose correspondence [5, 14]; so Corollary 17 contains the case $n = 4$ (Section 7 gives another short argument for it). In dimension 6 ($n \equiv 2 \pmod{4}$) both kinds occur: by the classification in Section 5, the component spaces of quadratic DU-4 permutations of $\mathbb{F}_2^6$ form exactly two classes, that of x^5 (Desarguesian kernel spread) and one with a non-Desarguesian kernel spread. For $n = 8$ non-Desarguesian kernel spreads are excluded as well, by Theorem 22.

5 Non-existence for $n = 8$

Theorem 22 (computer-assisted). *There is no quadratic differentially 4-uniform permutation of $\mathbb{F}_2^8$. More precisely, $\text{Alt}(8, \mathbb{F}_2)$ has no 8-dimensional subspace W such that*

- (i) *every nonzero element of W has rank 6, and*
- (ii) *every $a \in \mathbb{F}_2^8 \setminus \{0\}$ lies in the radicals of exactly three nonzero elements of W .*

Consequently every quadratic map $F : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^8$ all of whose derivative kernels have dimension 2 (equivalently, whose differential spectrum is $\{0, 4\}$) has a bent component.

For a subspace $U \subseteq \text{Alt}(n, \mathbb{F}_2)$ and $a \neq 0$ put $P_a(U) = \{w \in U : a \in \text{rad } w\}$, a subspace; (ii) says $\dim P_a(W) = 2$ for all $a \neq 0$.

Theorem 22 does not say that $\text{Alt}(8, \mathbb{F}_2)$ has no 8-dimensional subspace of constant rank 6. It excludes exactly the subspaces of constant rank 6 that also satisfy (ii), the condition imposed by the derivative kernels of a quadratic map with differential spectrum $\{0, 4\}$. Whether (i) alone can be satisfied is Conjecture H of Section 7, which remains open for $n = 8$.

Reduction. Let F be a quadratic DU-4 permutation of $\mathbb{F}_2^8$ with polar map β . By Theorem 5 every B_b ($b \neq 0$) has rank 6 and $s_a = 2$ for all $a \neq 0$. The map $b \mapsto B_b$ is linear and injective, so $W = \{B_b\}$ is 8-dimensional and satisfies (i). For $a \neq 0$ we have $B_b(a, \cdot) = 0$ if and only if $b \perp \text{Im } \beta(a, \cdot)$, so $\dim P_a(W) = 8 - \text{rank } \beta(a, \cdot) = s_a = 2$, which is (ii). For the last statement, if all $s_a = 2$ and no component is bent, all ranks are at most 6, and Lemma 1 gives $\sum_b (2^{8-\text{rank } B_b} - 1) = 3 \cdot 255$ with every term at least 3, so every component has rank 6 and $W = \{B_b\}$ again satisfies (i) and (ii).

We call W as in Theorem 22 a *solution*. In the rest of this section and in Appendix A, a *point* is a nonzero vector of $\mathbb{F}_2^8$; the letters a, r denote points and x, y, c, u, w denote forms. The theorem is proved by a search over subspaces U of a hypothetical solution, grown one dimension at a time. For a subspace U all of whose nonzero elements have rank 6, call a point a *full* for U if $\dim P_a(U) = 2$, and call a coset $x + U$ with $x \notin U$ *admissible* if all its elements have rank 6 and no element of $x + U$ has a full point of U in its radical. Let $\mathcal{C}(U)$ be the set of admissible cosets. Counting pairs (a, w) with $w \in U \setminus \{0\}$ and $a \in \text{rad } w \setminus \{0\}$ gives $\sum_{a \neq 0} (2^{\dim P_a(U)} - 1) = 3(2^d - 1)$ for $d = \dim U$; so if $d < 8$, some point a has $\dim P_a(U) \leq 1$.

Lemma 23 (radical closure). *Let W be a solution, $U \subseteq W$ a subspace of dimension $d < 8$ and $r \neq 0$ a point with $\dim P_r(U) \leq 1$. Then W contains an admissible coset $c + U$ that contains an element whose radical contains r . Moreover:*

- (P1) $\dim P_a(U) \leq 2$ for all $a \neq 0$;
- (P2) $|\mathcal{C}(U)| \geq 2^{8-d} - 1$;
- (P3) for every a with $\dim P_a(U) = 1$ (resp. 0), at least one (resp. three) admissible cosets contain an element whose radical contains a .

Proof. Since $\dim P_r(W) = 2 > \dim P_r(U)$, there is $c \in P_r(W) \setminus U$. The coset $c + U$ lies in $W \setminus \{0\}$, so its elements have rank 6; if some $c + u$ had a full point a of U in its radical, then $P_a(W) \supseteq P_a(U) + \langle c + u \rangle$ would have dimension 3. So $c + U$ is admissible, and $r \in \text{rad } c$. (P1) holds as $P_a(U) \subseteq P_a(W)$. The same argument shows that all $2^{8-d} - 1$ nonzero cosets of W/U are admissible, which is (P2). For (P3), the image of $P_a(W)$ in W/U has dimension $2 - \dim P_a(U)$; its nonzero elements are admissible cosets containing an element of $P_a(W)$. $\square$

The search. All forms of rank 6 on $\mathbb{F}_2^8$ are equivalent, so we may assume that a solution contains $J = e_0 \wedge e_1 + e_2 \wedge e_3 + e_4 \wedge e_5$ (radical $\langle e_6, e_7 \rangle$). Starting from $U_1 = \langle J \rangle$, a node U of dimension $d < 8$ is discarded if (P2) fails, or, for $d \geq 3$, if (P3) fails ((P1) holds for every node that is reached, by Lemma 28(b)); otherwise a point r with $\dim P_r(U) \leq 1$ is chosen by a fixed rule (for $d \leq 2$ the first point with $\dim P_r(U) = 1$, or with $\dim P_r(U) = 0$ if there is none; for $d \geq 3$ the point minimising the number of candidates below), and the children of U are the subspaces $U + \langle c \rangle$ for the admissible cosets $c + U$ containing an element whose radical contains r (the *candidates*), one for each orbit of a group $H \leq GL(8, 2)$ that stabilises U and fixes r . Here $g \in GL(8, 2)$ acts by $B^g(v, v') = B(gv, gv')$; it preserves ranks and maps solutions to solutions, since $\text{rad } B^g = g^{-1} \text{rad } B$. By Lemma 23, every solution containing U is mapped by an element of H to a solution containing one of the children, so every solution is equivalent to one containing a node of dimension 8; these are tested against (ii). The search is small because each step prescribes a point in the radical of the new element and because admissibility removes many cosets once full points appear (Appendix A.8).

Appendix A states the search as a procedure, proves its completeness (Theorem 29) and describes how the programs implement each step.

Symmetry. For U_1 we use explicit generators of subgroups of the stabiliser of J : the symplectic transvections $x \mapsto x + J(v, x)v$ for $v \in \langle e_0, \dots, e_5 \rangle$ of weight 1 or 2, the group $GL(2, 2)$ acting on $\langle e_6, e_7 \rangle$, and the maps $x \mapsto x + x_i e_6$, $x \mapsto x + x_i e_7$ ($i \leq 5$). Stabilisers of points and of cosets are generated by random products corrected by an orbit transversal and by random Schreier generators. A proper subgroup of a stabiliser only produces more children, so correctness does not depend on these subgroups being the full stabilisers; every generator of every group used is checked to stabilise U (and to fix r). For the two top levels the program writes a certificate: for each of the 740,736 candidates of U_1 and the 727,936 candidates of the three nodes of dimension 2, an explicit matrix of H mapping an orbit representative onto it.

Results. The search has 3 nodes of dimension 2, 486 of dimension 3, 1,466,318 of dimension 4 and 244,686,904 of dimension 5. Every node of dimension 5 fails (P2) (244,668,896 nodes) or (P3) (18,008 nodes), so no node of dimension 6 is reached and no solution exists. This proves Theorem 22. The run consisted of two parallel processes, which took 2.2 and 2.1 hours (4.3 CPU-hours in total) on a two-core virtual machine with 2.1 GHz Intel Xeon processors; a second run from the archived sources gave the same numbers at every dimension.

Verification.

- *A second implementation of the same search.* A second program (Python/numpy) was written separately from the mathematical specification of the search (including its tie-breaking rule), without access to the code of the first program. It belongs to the same study, so it is a separate code path rather than a third-party replication. It recomputes the table of rank-6 forms by Pfaffians (rank 6 if and only if the 8×8 Pfaffian vanishes and one of the 28 principal 6×6 Pfaffians does not) and obtains the same 149,920,960 forms as the first program, which uses Gaussian elimination. It checks the certificate of the two top levels line by line: each matrix is invertible, stabilises the node, fixes r and maps the representative onto the candidate; the listed candidates are exactly the candidates computed from the definitions; and the 486 nodes of dimension 3 are exactly those of the first program. It then searches the subtrees below the 486 nodes of dimension 3 without any symmetry reduction, so that no symmetry argument below dimension 3 is used. This search has 8,604,140 nodes of dimension 4 and 1,381,518,812 of dimension 5 (13.4 hours in total for two parallel processes); again every node of dimension

5 fails (P2) or (P3), and no solution exists. So the certificate of the two top levels and this search verify Theorem 22 without using the first program (Appendix A.6 lists what each part of the proof depends on). On the subtrees of five nodes of dimension 3 (from all three nodes of dimension 2), the first program run without symmetry below dimension 3 gives exactly the same numbers of nodes and of discarded nodes at every dimension. Agreement of the two programs tests the implementation; the pruning rules themselves are justified by Lemma 23.

- *Dimension 6.* The same program for $n = 6$ (rank 4, dimension 6) finds exactly two $GL(6, 2)$ -classes of solutions: the component space of x^5 (Desarguesian kernel spread, stabiliser of order $362,880 = |\Gamma L(3, 4)|$) and a class with a non-Desarguesian kernel spread in which 50 of the 210 pairs of kernel lines are closed (stabiliser of order 5,760). Both are component spaces of DU-4 permutations. An affine equivalence $G = A_2 \circ F \circ A_1$ changes the component space only through the linear part of A_1 (the output map A_2 only re-indexes the components, and affine terms do not change β), so each affine class of quadratic DU-4 permutations of $\mathbb{F}_2^6$ determines one of these two classes; the eight affine classes with $\delta = 4$ found in [8] are distributed over our two classes, both of which occur (one component space carries several permutations, which differ in their linear parts). Without symmetry reduction the program finds 12,288 solutions containing J ; this equals $|GL(6, 2)| \cdot 63 \cdot (1/362,880 + 1/5,760)/18,228 = 192 + 12,096$, the number predicted by the two stabilisers (18,228 is the number of forms of rank 4 in $\text{Alt}(6, \mathbb{F}_2)$). A plain enumeration that does not prescribe radical points (a depth-first search over all admissible cosets that reaches each subspace containing J exactly once, through its greedy basis modulo J) finds the same 12,288 subspaces. The same enumeration with the condition on full points dropped, so that a coset is only required to consist of forms of rank 4, again finds exactly these 12,288 subspaces. So for $n = 6$ condition (ii) follows from (i), and $\text{Alt}(6, \mathbb{F}_2)$ has exactly two $GL(6, 2)$ -classes of 6-dimensional subspaces of constant rank 4.

6 Local structure of kernel spreads and earlier approaches

For kernel lines $L \neq M$ let $k(L, M)$ be the number of kernel lines contained in $L \oplus M$, and call $\{L, M\}$ *closed* if $k(L, M) = 5$, i.e. $L \oplus M$ is a union of kernel lines. A Desarguesian kernel spread has $k(L, M) = 5$ for all pairs. Put $d(L, M) = \dim \langle \beta(L, M) \rangle$; since $\beta(L, L) = \beta(M, M) = 0$, this is the dimension of $\langle \beta(U, U) \rangle$ for $U = L \oplus M$.

Proposition 24. *Let $n \geq 6$ be even and let β have $s_a = 2$ for all $a \neq 0$ and all components of rank $n - 2$ (as for every quadratic DU-4 permutation, Theorem 5). For kernel lines $L \neq M$:*

- (a) $k(L, M) \in \{2, 3, 5\}$;
- (b) $d(L, M) = 4, 3, 2$ according as $k(L, M) = 2, 3, 5$;
- (c) if $n = 8$, no pair is closed, so $k(L, M) \in \{2, 3\}$.

Proof. (a) Let $U = L \oplus M$ and suppose U contains a third kernel line N . Three pairwise skew lines of $PG(3, 2)$ lie in a unique spread $\{L, M, N, P, Q\}$ of $PG(U)$; it is regular, so its lines are the $\mathbb{F}_4$ -points for some $\omega \in GL(U)$ with $\omega^2 + \omega + 1 = 0$. A line of U skew to L, M and N lies in the six points of $P \cup Q$, hence contains two points of P or of Q and equals P or Q ; so the kernel lines in U are among L, M, N, P, Q . The map $q(x) = \beta(x, \omega x)$ is quadratic on U and satisfies $q(\omega x) = \beta(\omega x, x + \omega x) = q(x)$, so it takes one value q_X on the nonzero points of each $\mathbb{F}_4$ -point X . Every component of q is a quadratic form on $\mathbb{F}_2^4$, which takes the value 1 an even number of times; hence $\sum_{x \in U} q(x) = 0$, i.e. $q_L + q_M + q_N + q_P + q_Q = 0$. A line $\mathbb{F}_4 x$ of U is a kernel line if and only if

$\beta(x, \omega x) = 0$ (as $s_x = 2$ and $\omega x \notin \{0, x\}$). So $q_L = q_M = q_N = 0$, hence $q_P = q_Q$, and P is a kernel line if and only if Q is. Thus $k \neq 4$.

(b) Let $U = L \oplus M$ and let N_0, N_2, N_4 be the numbers of $b \neq 0$ for which $B_b|_U$ has rank 0, 2, 4, so that $N_0 + N_2 + N_4 = 2^n - 1$; since $B_b|_U = 0$ exactly when $b \perp \langle \beta(U, U) \rangle$, we have $N_0 = 2^{n-d} - 1$. A point a on one of the k kernel lines inside U has $K_a \subseteq U$, so $\beta(a, \cdot)|_U$ has kernel K_a and $\dim \beta(a, U) = 2$; every other point $a \in U$ has $K_a \cap U = \langle a \rangle$ and $\dim \beta(a, U) = 3$. Remark 3 for U gives

$$15N_0 + 3N_2 = 3k(2^{n-2} - 1) + (15 - 3k)(2^{n-3} - 1). \quad (1)$$

If $k = 5$, then $\beta(x, \omega x) = 0$ on U , so by Remark 18, applied to each $B_b|_U$, every $B_b|_U$ has rank 0 or 4; thus $N_2 = 0$, and (1) gives $N_0 = 2^{n-2} - 1$, i.e. $d = 2$. If $k \in \{2, 3\}$, some point of U has $\dim \beta(a, U) = 3$, so $d \in \{3, 4\}$. If $k = 3$, then $v := q_P = q_Q \neq 0$ by (a), since P is not a kernel line; every $b \perp v$ has $B_b(x, \omega x) = 0$ on U , so $B_b|_U$ has rank 0 or 4 by Remark 18, and therefore $N_2 \leq (2^n - 1) - (2^{n-1} - 1) = 2^{n-1}$. Solving (1) for N_2 gives the following table.

k	d	N_0	N_2	$3N_0 + N_2$
2	3	$2^{n-3} - 1$	2^{n-2}	$3 \cdot 2^{n-2} - 3 - 2^{n-3}$
2	4	$2^{n-4} - 1$	$9 \cdot 2^{n-4}$	$3 \cdot 2^{n-2} - 3$
3	3	$2^{n-3} - 1$	$3 \cdot 2^{n-3}$	$3 \cdot 2^{n-2} - 3$
3	4	$2^{n-4} - 1$	$11 \cdot 2^{n-4} > 2^{n-1}$	excluded
5	2	$2^{n-2} - 1$	0	$3 \cdot 2^{n-2} - 3$

So every pair has $3N_0 + N_2 \leq 3 \cdot 2^{n-2} - 3$, with equality unless $(k, d) = (2, 3)$. We show that the average of $3N_0 + N_2$ over all unordered pairs of kernel lines equals $3 \cdot 2^{n-2} - 3$; then no pair has $(k, d) = (2, 3)$, which proves (b).

Fix $b \neq 0$. The ordered pairs $(x, y) \in V^2$ with $B_b(x, y) = 0$ number $4 \cdot 2^n + (2^n - 4)2^{n-1}$: for the 4 vectors $x \in R_b$ every y qualifies, and for the other x the functional $B_b(x, \cdot)$ is nonzero and vanishes on 2^{n-1} vectors. Of these pairs, $2^{n+1} - 1$ have $x = 0$ or $y = 0$, and $3(2^n - 1)$ consist of nonzero vectors in a common kernel line (B_b vanishes on $L' \times L'$ for every kernel line L' , and there are $|\Sigma| = (2^n - 1)/3$ kernel lines with 9 such ordered pairs each). This leaves

$$Z = 2^{2n-1} - 3 \cdot 2^n + 4$$

ordered pairs of nonzero vectors in distinct kernel lines. Group them by the ordered pair (L', M') of lines. Let $\rho \in \{0, 1, 2\}$ be the rank of the 2×2 matrix of B_b on $L' \times M'$; the number of $(x, y) \in (L' \setminus 0) \times (M' \setminus 0)$ with $B_b(x, y) = 0$ is 9, 5, 3 for $\rho = 0, 1, 2$ (for $\rho = 1$, $B_b(x, y) = \lambda(x)\mu(y)$ with nonzero functionals λ, μ , and there are $1 \cdot 3 + 3 \cdot 1 - 1$ zeros; for $\rho = 2$ every x has exactly one partner y). With $n_\rho(b)$ the number of ordered pairs of distinct kernel lines of type ρ , we get $n_0 + n_1 + n_2 = |\Sigma|(|\Sigma| - 1)$ and $9n_0 + 5n_1 + 3n_2 = Z$, hence

$$3n_0(b) + n_1(b) = \frac{1}{2}(Z - 3|\Sigma|(|\Sigma| - 1)) = \frac{1}{6}(2^n - 4)(2^{n-1} - 2).$$

Since B_b vanishes on $L' \times L'$ and on $M' \times M'$, its rank on $U' = L' \oplus M'$ is 2ρ ; so type 0 means $B_b|_{U'} = 0$ and type 1 means that $B_b|_{U'}$ has rank 2. Each unordered pair $\{L', M'\}$ gives two ordered pairs of the same type, so summing over $b \neq 0$ and exchanging the order of summation,

$$\sum_{b \neq 0} (3n_0(b) + n_1(b)) = 2 \sum_{\{L', M'\}} (3N_0(U') + N_2(U')).$$

The left side is $(2^n - 1)(2^n - 4)(2^{n-1} - 2)/6$ and there are $\binom{|\Sigma|}{2} = (2^n - 1)(2^n - 4)/18$ unordered pairs, so the average of $3N_0 + N_2$ is $\frac{(2^n - 1)(2^n - 4)(2^{n-1} - 2)/12}{(2^n - 1)(2^n - 4)/18} = \frac{3}{2}(2^{n-1} - 2) = 3 \cdot 2^{n-2} - 3$.

(c) Let $n = 8$ and $k = 5$. By (b), $S = \langle \beta(U, U) \rangle$ has dimension 2, so there are 63 values $b \neq 0$ with $b \perp S$, i.e. with $B_b|_U = 0$. For each of them $U \cap R_b \neq 0$: otherwise $U \oplus R_b$ would be a totally isotropic subspace of dimension 6 for B_b , whereas a totally isotropic subspace for a form of rank 6 on $\mathbb{F}_2^8$ has dimension at most $2 + 3 = 5$ (the radical has dimension 2, and a totally isotropic subspace of the nondegenerate quotient of dimension 6 has dimension at most 3). So there are at least 63 pairs (a, b) with $a \in U \setminus \{0\}$, $b \perp S$, $b \neq 0$ and $a \in R_b$. But every $a \neq 0$ lies in exactly three radicals (those of the $b \in (\text{Im } \beta(a, \cdot))^\perp$), so there are at most $3 \cdot 15 = 45$ such pairs, a contradiction. $\square$

Remark 25. For $n = 6$ the argument of (c) forces $R_b \subseteq U$ for the 15 values of b , i.e. exactly 45 pairs, and closed pairs do occur: by the classification for $n = 6$ in Section 5, 50 or all 210 of the pairs are closed, the other pairs having $k = 2$ and $d = 4$. Part (c) gives a second proof of Corollary 17 for $n = 8$ (it shares the $\mathbb{F}_4$ -trace step of Remark 18). For $n = 8$ the hypotheses of Proposition 24 cannot be satisfied, by Theorem 22; parts (a) and (b) hold in every even dimension $n \geq 6$.

Earlier approaches for $n = 8$. Before the search of Section 5 we tried three other approaches; they agree with Theorem 22 but did not settle the case. Programs, logs and full proofs are in the repository.

- *Structured classes.* Representing $\mathbb{F}_{2^8}$ with the modulus $X^8 + X^4 + X^3 + X^2 + 1$ and using exponents $2^i + 2^j$, no binomial $x^{d_1} + c x^{d_2}$ (96,390 polynomials), no polynomial with coefficients in $\mathbb{F}_2$ (2^{28} polynomials) and no trinomial (213,021,900 polynomials) has all 255 components of rank 6; the closest are 12,288 polynomials with coefficients in $\mathbb{F}_2$ that have three components of other ranks.
- *SAT.* A direct encoding of quadratic DU-4 permutations (variables $\beta(e_i, e_j)$ and $F(e_i)$, symmetry breaking by Lemma 11) is unsatisfiable for $n = 4$ (with a DRAT proof checked by `drat-trim` [23]) and satisfiable for $n = 6$, but `kissat` [15] gave no verdict for $n = 8$ within several hours, also after splitting by the local structure of Proposition 24. With the Desarguesian condition added, the instance is unsatisfiable within a second (a DRAT-checked consistency check of Corollary 17).
- *Symmetries.* Call $(A, B) \neq (I, I)$ in $GL(8, 2)^2$ a symmetry of β if $\beta(Ax, Ay) = B\beta(x, y)$; every nontrivial affine self-equivalence of F gives one. By a case analysis over the conjugacy classes of the possible pairs (A, B) with A of prime order, using SAT calls certified by DRAT proofs and exhaustive enumerations, we had excluded every β with a symmetry (the proof and the computations are in the repository). This statement is now contained in Theorem 22.

7 Constant-rank subspaces and an open problem

For $n \geq 4$, Theorem 5 says that the component forms of a quadratic DU-4 permutation span an n -dimensional subspace W of $\text{Alt}(n, \mathbb{F}_2)$ all of whose nonzero elements have rank $n - 2$ (the map $b \mapsto B_b$ is injective, since a nonzero b gives a component of rank $n - 2 > 0$). For $n = 4$ no such subspace exists: the forms of rank ≤ 2 in $\text{Alt}(4, \mathbb{F}_2)$ form the Klein quadric $Q^+(5, 2)$, whose largest totally singular subspaces have dimension 3 (equivalently, the MacWilliams transform in the alternating-forms scheme [9] has a negative coefficient). For $n = 8$ the transform is non-negative and integral, so it gives no obstruction (Table 1).

Constant-rank subspaces of alternating forms have mostly been studied over larger fields. Gow and Quinlan [12] construct them from Galois extensions; as far as we can see, these constructions do not give n -dimensional subspaces of constant rank $n - 2$ when $4 \mid n$ (for instance the component

space of x^5 on $\mathbb{F}_{2^8}$ contains forms of ranks 4 and 8). The extension of these constructions by Gupta and Mandal [13] assumes that the characteristic is not two. Gow [11] shows that if $q \geq m + 1$, the distinct radicals of the nonzero elements of an n -dimensional subspace of $\text{Alt}(n, \mathbb{F}_q)$ of constant rank m form a spread [11, Theorem 13], and he bounds the dimension of constant-rank subspaces under the same field-size hypothesis [11, Theorem 11]. These results do not apply here, since $q = 2$ and $m = n - 2$, and over $\mathbb{F}_2$ the spread conclusion indeed fails: for $n = 6$, the non-Desarguesian class of Section 5 is a 6-dimensional subspace of constant rank 4 whose 63 nonzero elements have 61 distinct radicals, so the radicals do not form a spread (for x^5 they are the 21 lines of the kernel spread, each occurring three times). In our setting the spread comes from the derivative kernels, that is, from the condition $s_a = 2$, and not from the radicals. We are not aware of results settling the existence question over $\mathbb{F}_2$.

In Table 1, $W^\perp$ is taken with respect to the pairing $\langle A, A' \rangle = \sum_{i < j} A_{ij} A'_{ij}$. A rank-2 element of $\text{Alt}(n, \mathbb{F}_2)$ is $u \wedge v$ (the form $(x, y) \mapsto \langle u, x \rangle \langle v, y \rangle + \langle v, x \rangle \langle u, y \rangle$) for a unique line $\langle u, v \rangle$, and $\langle A, u \wedge v \rangle = A(u, v)$. Hence the rank-2 elements of $W^\perp$ correspond to the lines $\langle u, v \rangle$ with $B(u, v) = 0$ for all $B \in W$, i.e. with $\beta(u, v) = 0$. Counting ordered pairs (u, v) with $v \in K_u \setminus \{0, u\}$ and using Lemma 1, the number of these lines is $\sum_a (2^{s_a} - 2)/6 = (2^n - 1)/3$ for every n -dimensional subspace of constant rank $n - 2$, in agreement with the entry A'_1 of Table 1. These lines form a spread if and only if $s_a = 2$ for all a . This holds for DU-4 permutations (Theorem 5); we do not know whether it follows from the constant-rank condition in general. For $n = 6$ it does, by the enumeration at the end of Section 5.

n	d	verdict	dual rank distribution $A'_0, \dots, A'_{n/2}$
4	4	impossible	1, 5, -2
6	6	consistent	1, 21, 210, 280
8	8	consistent	1, 85, 20706, 590920, 436864

Table 1: MacWilliams transform of the rank distribution $(1, 0, \dots, 0, 2^d - 1, 0)$ of a d -dimensional subspace of $\text{Alt}(n, \mathbb{F}_2)$ of constant rank $n - 2$; A'_r is the number of elements of rank $2r$ in $W^\perp$.

Conjecture 26 (H). *For $n \equiv 0 \pmod{4}$, $\text{Alt}(n, \mathbb{F}_2)$ has no n -dimensional subspace whose nonzero elements all have rank $n - 2$.*

Conjecture H would imply that no quadratic DU-4 permutation exists when $4 \mid n$, but it is a stronger statement: a constant-rank subspace is only a necessary ingredient. A permutation additionally needs $s_a = 2$ for all a and, for a quadratic map Q with polar map β , a linear map Λ with $Q(a) + \Lambda(a) \notin \text{Im } \beta(a, \cdot)$ for all $a \neq 0$. A counterexample to Conjecture H would therefore not by itself produce a permutation. For $n = 8$, Theorem 22 proves Conjecture H for subspaces with $s_a = 2$ for all a (in the notation of Section 5, $\dim P_a(W) = 2$). For a general 8-dimensional subspace of constant rank 6 the counting argument above gives $\sum_{a \neq 0} (2^{\dim P_a(W)} - 1) = 3 \cdot 255$ with $\dim P_a(W) \geq 1$, so either all $\dim P_a(W) = 2$, which is excluded, or some point lies in the radicals of at least seven elements; the second case is not covered by our search, and Conjecture H remains open for $n = 8$. (For $n = 6$ the analogous second case does not occur, as noted above.)

8 Conclusion

By [7], a quadratic differentially 4-uniform permutation of $\mathbb{F}_2^n$ has all components of rank $n - 2$ and all derivative kernels of dimension 2. We observed that these kernels form a line spread that is totally

isotropic for every component, and showed that when $4 \mid n$ this spread cannot be Desarguesian; the same argument shows in general that a Desarguesian kernel t -spread forces n/t to be odd. This explains why no $\mathbb{F}_4$ -quadratic construction, for instance a Gold map with an added linear part, works when $4 \mid n$, gives a uniform reason for the parity hypothesis in the binomial family of [4], and places the classical non-existence of quadratic APN permutations in even dimension in the same framework. For $n = 8$ we showed, by an exhaustive search over the possible spaces of components, that no quadratic DU-4 permutation exists (DU-4 permutations of higher degree do exist, the inverse map being the best-known example); in fact the search never gets beyond five of the eight dimensions. Together with $n = 4$, this suggests that quadratic DU-4 permutations of $\mathbb{F}_2^n$, n even, exist only for $n \equiv 2 \pmod{4}$, where the Gold maps provide them for every $n \geq 6$. The next open case is $n = 12$, which is far out of reach of the same search. Theorem 22 excludes only the constant-rank subspaces that satisfy the radical condition (ii); Conjecture H, which drops this condition, is open already for $n = 8$.

Reproducibility

The programs, logs and certificates are available in the repository

<https://github.com/AlexGuseinov/quadratic-perm-4n>.

Its file `ARTIFACT.md` lists, for each computational statement of this paper, the command that reproduces it, the expected output and what has to be trusted; `expected_counts.json` contains the numbers of nodes of both programs in machine-readable form, and the checking scripts compare their logs with it.

First program (directory `search_n8/`). The search of Section 5 is `crsub.enum.c` (C, compiled with `gcc 13.3 -O3 -march=native`; it needs an x86-64 processor with the BMI2 instructions). The script `run_n8.sh` runs it in two parallel parts and records the compiler, the command lines and the SHA-256 hashes of the sources, the binary and the table of rank-6 forms; `check_n8_logs.py` checks that all 486 nodes of dimension 3 were processed, that no node of dimension 6 occurred and that the numbers of nodes are the expected ones. All random choices in the computation of stabilisers are derived from a fixed seed and the path from the root, so the program is deterministic and a rerun reproduces the same tree and the same numbers. The reported run used Ubuntu 24.04 on a two-core virtual machine with Intel Xeon processors at 2.1 GHz: two parallel processes of 2.2 and 2.1 hours (4.3 CPU-hours), at most 0.3 GB of memory per process. The second run, from the archived sources, took 3.4 and 3.3 hours because it shared the machine with the second program.

Second program. The directory `second_impl/` contains the second implementation (Python 3.11 with NumPy 2.4), the certificate of the two top levels (`cert8.txt.xz`, 1,468,672 lines, with its SHA-256 hash), its checker and the logs of the complete run over the 486 nodes of dimension 3 (two parallel processes of 6.8 and 6.7 hours, 13.4 hours in total, partly on the shared machine; at most 0.6 GB of memory per process). One command, `sh second_impl/run_all.sh`, carries out the verification of Theorem 22 that does not use the first program: it computes the table of rank-6 forms by Pfaffians, checks the certificate, searches the 486 subtrees and compares the result with the expected numbers. The only data it takes from the first program are the certificate and the list of the 486 nodes of dimension 3, and it checks both against the definitions; the checker shares no code with the program that wrote the certificate.

Small computations. The script `check_all.sh` re-checks, in about a minute, the small computations only (rank counts, the MacWilliams table, the SAT model for $n = 4$ and $n = 6$, the Desarguesian instance for $n = 8$, and the examples for x^5); it does not run the search for $n = 8$.

The enumerations for $n = 6$ of Section 5, the searches over structured classes, the symmetric-case computations and the long solver runs mentioned in Section 6 are documented in the repository by their commands, logs and certificates.

A The computation in mathematical terms

This appendix describes the search of Section 5 in mathematical terms: the objects it works with, the procedure and a proof that it is complete, how each quantity is computed, how the symmetry reduction is certified, and what each part of the proof of Theorem 22 depends on. Throughout $n = 8$, $V = \mathbb{F}_2^8$, and forms are elements of $\text{Alt}(8, \mathbb{F}_2)$.

A.1 The objects

A form B is determined by its 28 values $B(e_i, e_j)$, $i < j$; this identifies $\text{Alt}(8, \mathbb{F}_2)$ with $\mathbb{F}_2^{28}$, and subspaces of forms with subspaces of $\mathbb{F}_2^{28}$. In the programs and their logs the pairs are ordered lexicographically, $(0, 1), (0, 2), \dots, (6, 7)$, a form is written as the integer whose k -th binary digit is its k -th coordinate, and a point a as the integer $\sum_i a_i 2^i$; thus $J = 2^0 + 2^{13} + 2^{22} = 4,202,497$. A table of all 2^{28} forms records which of them have rank 6. The first program computes ranks by Gaussian elimination; the second uses Pfaffians: $\text{rank } B = 8$ if and only if the Pfaffian of B is nonzero, and $\text{rank } B \geq 6$ if and only if one of the 28 principal 6×6 Pfaffians is nonzero. Both give the same 149,920,960 forms of rank 6, the number given by the classical formula for the number of alternating forms of given rank (see [9]).

If the basis of a subspace U is in reduced echelon form (each basis vector has a pivot coordinate in which the other basis vectors vanish), every coset $y + U$ contains exactly one vector that vanishes in all pivot coordinates; it is obtained from y by adding the basis vectors whose pivot coordinate is 1 in y . Sets of cosets are stored as sets of these representatives.

For $a \in V$ let $\varphi_a : \text{Alt}(8, \mathbb{F}_2) \rightarrow V^*$, $\varphi_a(B) = B(a, \cdot)$, that is, $\varphi_a(B)(e_k) = \sum_i a_i B(e_i, e_k)$; it is linear (an 8×28 matrix over $\mathbb{F}_2$), and $a \in \text{rad } B$ if and only if $\varphi_a(B) = 0$. For a subspace U we have $P_a(U) = U \cap \ker \varphi_a$, so

$$\dim P_a(U) = \dim U - \dim \varphi_a(U).$$

A *node* is a subspace U all of whose nonzero elements have rank 6 and with $\dim P_a(U) \leq 2$ for all $a \neq 0$. For a node U let $\mathcal{F}(U) = \{a \neq 0 : \dim P_a(U) = 2\}$ (the full points). A coset $x + U$, $x \notin U$, is admissible if all its elements have rank 6 and $\varphi_a(x) \notin \varphi_a(U)$ for every $a \in \mathcal{F}(U)$ (by the display, this says that no element of $x + U$ has a in its radical); $\mathcal{C}(U)$ is the set of admissible cosets. For a point r put $\text{Cand}(U, r) = \{x + U \in \mathcal{C}(U) : \varphi_r(x) \in \varphi_r(U)\}$. For $g \in GL(V)$ and a form B put $B^g(v, v') = B(gv, gv')$ for $v, v' \in V$, and $U^g = \{u^g : u \in U\}$.

A.2 Two lemmas

Lemma 27 (group action). *Let $g, h \in GL(V)$, B a form, U a subspace and $a \in V$.*

- (a) $B \mapsto B^g$ is linear and bijective, $(B^g)^h = B^{gh}$ and $\text{rank } B^g = \text{rank } B$.
- (b) $a \in \text{rad } B^g$ if and only if $ga \in \text{rad } B$; hence $P_a(U^g) = P_{ga}(U)^g$.
- (c) U^g is a node if and only if U is, and then $\mathcal{F}(U^g) = g^{-1}\mathcal{F}(U)$. If W is a solution, so is W^g .
- (d) If U is a node, $U^g = U$ and $gr = r$, then $x + U \mapsto x^g + U$ is a permutation of $\mathcal{C}(U)$ and of $\text{Cand}(U, r)$.

Proof. (a) In matrix form $B^g = g^\top Bg$. (b) $B^g(a, v) = B(ga, gv)$ vanishes for all $v \in V$ if and only if $B(ga, \cdot) = 0$. Hence $w^g \in P_a(U^g)$ if and only if $w \in P_{ga}(U)$. (c) follows from (a) and (b), since $\dim P_a(U^g) = \dim P_{ga}(U)$. (d) By (c), $g\mathcal{F}(U) = \mathcal{F}(U)$. For $y \in x + U$, y^g has rank 6 if and only if y has, and by (b) y^g has a point a of $\mathcal{F}(U)$ in its radical if and only if y has the point $ga \in \mathcal{F}(U)$ in its radical; likewise $r \in \text{rad } y^g$ if and only if $r = gr \in \text{rad } y$. So $x + U$ and $x^g + U = (x + U)^g$ are admissible (resp. candidates) together; the inverse map is given by g^{-1} . $\square$

Lemma 28 (one step). *Let U be a node of dimension d , $x + U \in \mathcal{C}(U)$ and $U' = U + \langle x \rangle$. For $a \neq 0$ put $\varepsilon_a = 1$ if $\varphi_a(x) \in \varphi_a(U)$ and $\varepsilon_a = 0$ otherwise.*

- (a) $\dim P_a(U') = \dim P_a(U) + \varepsilon_a$ for all $a \neq 0$.
- (b) $\varepsilon_a = 0$ for $a \in \mathcal{F}(U)$. Hence U' is a node and $\mathcal{F}(U) \subseteq \mathcal{F}(U')$.
- (c) For $y \notin U'$, the coset $y + U'$ is admissible for U' if and only if $y + U$ and $y + x + U$ are admissible for U and $\varphi_a(y) \notin \varphi_a(U')$ for all $a \in \mathcal{F}(U') \setminus \mathcal{F}(U)$.
- (d) $x + U$ contains an element with r in its radical if and only if $\varphi_r(x) \in \varphi_r(U)$.
- (e) If $d < 8$, some a has $\dim P_a(U) \leq 1$.

Proof. (a) $\varphi_a(U') = \varphi_a(U) + \langle \varphi_a(x) \rangle$ has dimension $\dim \varphi_a(U) + 1 - \varepsilon_a$, and $\dim U' = d + 1$. (b) The first statement is the definition of admissibility. So $\dim P_a(U')$ stays 2 on $\mathcal{F}(U)$ and is at most $1 + 1$ elsewhere; the elements of $U' \setminus U = x + U$ have rank 6. (c) $y + U'$ is the union of the two cosets $y + U$ and $y + x + U$, both different from U . All elements of $y + U'$ have rank 6 if and only if this holds for both cosets. For $a \in \mathcal{F}(U)$ the condition $0 \notin \varphi_a(y + U')$ holds if and only if it holds for both cosets, and for $a \in \mathcal{F}(U') \setminus \mathcal{F}(U)$ it is the condition in the statement; by (b) this covers $\mathcal{F}(U')$. (d) $\varphi_r(x + u) = \varphi_r(x) + \varphi_r(u)$. (e) Each nonzero element of U has exactly three nonzero points in its radical, so $\sum_{a \neq 0} (2^{\dim P_a(U)} - 1) = 3(2^d - 1) < 3 \cdot 255$; if all $\dim P_a(U)$ were 2, the sum would be $3 \cdot 255$. $\square$

A.3 The procedure and its completeness

$\text{Search}(U, H)$ takes a node U of dimension d and a group $H \leq GL(V)$ with $U^h = U$ for all $h \in H$.

1. If $d = 8$: report U if $\dim P_a(U) = 2$ for all $a \neq 0$, and return.
2. If $|\mathcal{C}(U)| < 2^{8-d} - 1$, return. (P2)
3. If $d \geq 3$ and some a with $\dim P_a(U) = 1$ (resp. 0) has fewer than one (resp. three) cosets in $\text{Cand}(U, a)$, return. (P3)
4. Choose r with $\dim P_r(U) \leq 1$ (it exists by Lemma 28(e)). For $d \leq 2$, r is the smallest point, as an integer $\sum_i a_i 2^i$, with $\dim P_r(U) = 1$, or with $\dim P_r(U) = 0$ if there is none; for $d \geq 3$, r minimises $4|\text{Cand}(U, a)| + [\dim P_a(U) = 0]$ over the points with $\dim P_a(U) \leq 1$, ties going to the smallest point. Let H_r be a subgroup of $\{h \in H : hr = r\}$.
5. Let $c_1 + U, \dots, c_m + U$ be representatives of the H_r -orbits on $\text{Cand}(U, r)$ (Lemma 27(d)). For each i let H_i be a subgroup of the stabiliser of $c_i + U$ in H_r , and call $\text{Search}(U + \langle c_i \rangle, H_i)$.

The search is $\text{Search}(\langle J \rangle, H_1)$, where H_1 is the group generated by the elements listed in Section 5, each of which fixes J . By Lemma 28(b) every call is made with a node, and H_i stabilises $U + \langle c_i \rangle$ because it stabilises U and $c_i + U$. The choice of r and of the subgroups H_r, H_i does not affect correctness; only the orbits and the tree size depend on them.

Theorem 29. *If a solution W exists, then the search reports a node U with $U = W^g$ for some $g \in GL(V)$.*

Proof. We show by induction on d that there is a call $\text{Search}(U_d, H_d)$ with $\dim U_d = d$ and an element g_d with $U_d \subseteq W^{g_d}$. For $d = 1$, W contains a form of rank 6, which is J^g for some g (all forms of rank 6 are equivalent); then $J \in W^{g^{-1}}$. Let $d < 8$, $U = U_d$ and $W' = W^{g_d}$, a solution by Lemma 27(c). By Lemma 23, steps 2 and 3 do not return, and it gives $c \in W'$ with $c + U \in \text{Cand}(U, r)$. Let $c_i + U$ be the representative of its H_r -orbit and $h \in H_r$ with $(c + U)^h = c_i + U$. As $U^h = U$, $U + \langle c_i \rangle = (U + \langle c \rangle)^h \subseteq W'^h = W^{g_d h}$, so $U_{d+1} = U + \langle c_i \rangle$ and $g_{d+1} = g_d h$ satisfy the claim. For $d = 8$, $U_8 \subseteq W^{g_8}$ and both have dimension 8, so $U_8 = W^{g_8}$, which satisfies (ii) and is reported in step 1. $\square$

Since the search reports nothing, Theorem 22 follows. Of the programs, the proof needs only that they carry out steps 1–5 as stated: that $\mathcal{C}(U)$, the dimensions $\dim P_a(U)$ and $\text{Cand}(U, r)$ are computed correctly, that every group element used stabilises U and fixes r , and that the representatives meet every orbit.

A.4 How each step is computed

1. *Start.* $\mathcal{C}(U_1)$, $U_1 = \langle J \rangle$, is obtained by testing all 2^{27} cosets of $\langle J \rangle$ (both elements must have rank 6; there are no full points yet). It has 41,891,328 elements.
2. *Children.* When $x + U$ is added, Lemma 28(a) updates $\dim P_a$ for all 255 points: for each a one tests whether $\varphi_a(x) \in \varphi_a(U)$, using an echelon basis of $\varphi_a(U) \subseteq V^*$ that is kept for every point. Then $\mathcal{C}(U')$ is computed by Lemma 28(c): for each representative y of a coset in $\mathcal{C}(U)$, the representative of $y + x + U$ is looked up in $\mathcal{C}(U)$; if it is there, $y + U'$ is admissible unless $\varphi_a(y) \in \varphi_a(U')$ for a new full point a .
3. *Steps 2–4.* $|\mathcal{C}(U)|$ is the size of this set, and for each point a with $\dim P_a(U) \leq 1$ the number $|\text{Cand}(U, a)|$ is the number of representatives y with $\varphi_a(y) \in \varphi_a(U)$ (Lemma 28(d)).
4. *Step 5.* The orbits of H_r on $\text{Cand}(U, r)$ are found by breadth-first search: the generators of H_r act linearly on $\mathbb{F}_2^{28}$ ($B^g = g^T B g$), and for each candidate the search records a product u of generators mapping the orbit representative onto it; the representative of an orbit is its first element in the order in which the candidates are stored. Generators of H_r and of H_i are obtained by composing a random product g of generators of H with the inverse of the recorded element u that has the same image as g , and as Schreier generators. Every generator is checked to stabilise U and to fix r (for H_r) or the coset $c_i + U$ (for H_i); then so does every element of the group they generate.
5. *Order of work.* The tree is traversed depth first. The random choices in step 5 are determined by the path from the root, so the 486 subtrees below the nodes of dimension 3 can be processed separately and distributed over several processes. The numbers of nodes of the first program at dimensions 4 and 5 depend on the subgroups found in step 5, hence on these choices; those of the second program depend only on the 486 nodes of dimension 3.

In the first program (the file `crsub_enum.c` in the directory `search_n8/`), the function `points_child` implements Lemma 28(a),(b), the function `make_child` implements (c), the function `choose_point` implements steps 1–4, and the function `expand` implements step 5. The second program carries out the same steps in Python with `numpy`; it was written separately, from a specification of steps 1–5, and shares no code with the first.

A.5 Symmetry and the certificate

The symmetry reduction matters most at the top: the 740,736 candidates of U_1 fall into 3 orbits, and the 727,936 candidates of the three nodes of dimension 2 into 486 orbits. For these four nodes the first program writes a certificate: for every candidate $x+U$, the index o of its orbit representative c_o and a matrix $u \in GL(8, 2)$ with $U^u = U$, $ur = r$ and $(c_o + U)^u = x + U$. Checking it requires only linear algebra: check that r is the point prescribed by step 4 (in particular $\dim P_r(U) \leq 1$); compute $\text{Cand}(U, r)$ from the definitions and compare it with the listed cosets; for each line, check that u is invertible, maps a basis of U into U , fixes r and maps c_o into $x + U$; and check that every $c_o + U$ is a candidate and that the nodes of the next dimension are the subspaces $U + \langle c_o \rangle$. Then, as in the proof of Theorem 29, every solution containing U is equivalent to one containing some $U + \langle c_o \rangle$. Below dimension 3 the second program uses no symmetry at all.

A.6 What the proof depends on

Ingredient	How it is established
Reduction to (i) and (ii)	Theorem 5 and Section 5 (proved)
Lemmas 23, 27, 28 and Theorem 29	proved
Table of forms of rank 6	computed twice (Gaussian elimination, Pfaffians), identical; the count agrees with the closed formula
Reduction to the 486 nodes of dimension 3	certificate of 1,468,672 matrices, checked by the second program
No node of dimension 6 below them	first program (with symmetry; two complete runs from the archived sources) and second program (without symmetry; one complete run)

To verify Theorem 22 without the first program, one needs the proved statements, one of the two rank tables, the certificate check and the search of the second program. The last item cannot be replaced by a short certificate: it visits more than 10^9 nodes.

A.7 A small example

For $n = 6$ the same procedure (rank 4, dimension 6, $J = e_0 \wedge e_1 + e_2 \wedge e_3$ with radical $\langle e_4, e_5 \rangle$) gives a tree that can be followed by hand. $\mathcal{C}(U_1)$ has 5,086 elements. Step 4 chooses $r = e_4$, the smallest point with $\dim P_r(U_1) = 1$; 366 cosets are candidates and fall into 2 orbits, giving 2 nodes of dimension 2. These have 144 candidates in 6 orbits. Of the 6 nodes of dimension 3, two fail (P3) and the others have 32 candidates in 15 orbits. Of the 15 nodes of dimension 4, six fail (P2) and the others have 27 candidates, each forming its own orbit. Each of the 27 nodes of dimension 5 has exactly one candidate, and the 27 nodes of dimension 6 all satisfy (ii). They are 9 distinct subspaces, in the two $GL(6, 2)$ -classes of Section 5. Without symmetry reduction the procedure gives 12,288 subspaces containing J , the same set as the plain enumeration of Section 5, which does not prescribe radical points.

A.8 Why the search is small

Let $p = 149,920,960/2^{28} \approx 0.5585$ be the proportion of forms of rank 6. If the ranks of the 2^d elements of a coset of a d -dimensional subspace behaved independently, there would be about $2^{28-d}p^{2^d}$ cosets consisting of forms of rank 6: about $4.2 \cdot 10^7$, $6.5 \cdot 10^6$, $3.2 \cdot 10^5$, $1.5 \cdot 10^3$ and 0.07 for $d = 1, \dots, 5$. For four random subspaces of constant rank 6 of each dimension the exact numbers

are within 2% of this for $d \leq 3$, about twice as large for $d = 4$ (about 3,200), and between 6 and 8 for $d = 5$; no random 6-dimensional subspace had any such coset. A candidate must moreover contain an element with the chosen point r in its radical: $\varphi_r(x)$ must lie in the subspace $\varphi_r(U)$, of dimension $d - \dim P_r(U)$, of the 7-dimensional space of linear forms vanishing at r , a condition of probability about $2^{d - \dim P_r(U) - 7}$; and admissibility removes further cosets once full points appear. In the search of the second program, which uses no symmetry there, the nodes of dimension 3 have about 17,700 candidates on average and the nodes of dimension 4 about 160, and every node of dimension 5 has fewer than 7 admissible cosets or fails (P3). This explains why the search ends at dimension 5; it is an explanation, not part of the proof.

A.9 Counts

Numbers of nodes by dimension; in parentheses the nodes returned in step 2 and in step 3. The second program takes the nodes of dimension at most 3 from the certificate and uses no symmetry reduction below them. In both programs no node of dimension at most 4 is returned in step 2 or 3.

dimension	first program	second program
1	1	1 (certificate)
2	3	3 (certificate)
3	486	486
4	1,466,318	8,604,140
5	244,686,904 (244,668,896; 18,008)	1,381,518,812 (1,381,416,881; 101,931)
6	0	0

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